

Full-pair RNA alignment comparison

Caris TN26-279853 | Fixed-cohort research QC | 5 October 2026

Five of the six stronger DNA/RNA discrepancy positions persist under the second alignment pipeline, including both linked PABPC1 sites. The APOD discrepancy is absent in the new counts and is demonstrably alignment-dependent. The experiment does not establish which placement is biologically correct or change the existing treatment-related research priorities.

Same starting cohort; coverage losses retained

All 23,209,264 original RNA read pairs were processed against the resident full primary-genome reference. The comparison was frozen at 4,015 originally jointly callable public SNPs, with the original DNA counts and categories unchanged. The new RNA remains callable at depth ≥ 20 at 3,932 sites; 83 fall below 20, and none loses all A/C/G/T evidence.

Among the same 2,873 DNA-dominant sites	Original RNA	New RNA
Depth ≥ 20 and same allele $\geq 90\%$	2,860 / 2,873	2,805 / 2,873
Depth ≥ 20 and same allele $\geq 98\%$	2,760 / 2,873	2,721 / 2,873
Depth below 20 (kept in denominator)	0	59
Depth ≥ 20 but same allele $< 90\%$	13	9

Of the original 13 exceptions, five reach $\geq 90\%$ same-allele retention and eight remain below it. One additional, low-depth MLLT3 site crosses below 90%. The 59 coverage failures came from previously passing sites. No baseline DNA-dominant site has a callable $\geq 98\%$ opposite REF/ALT allele. Fraction-only retention that ignores minimum depth is a separate statistic and is not substituted for the table above.

Six stronger original discrepancies

Locus (GRCh38, 1-based)	Original DNA	Original RNA	New RNA
COL5A2 chr2:189043211	G58	A116, G327	A116, G326
APOD chr3:195568834	G87	C27, G36	G86
EXOC3 chr5:443259	C153	A22, C74, G1	A23, C35, G1
MSH3 / DHFR chr5:80654917	C912	C9, G104	C9, G65
PABPC1 chr8:100705591	A94	A400, G65	A409, G61
PABPC1 chr8:100705604	G101	A1, C78, G358	A1, C73, G356

Counts are unambiguous query-name units, not unique molecules. The strong screen requires both DNA/RNA depths ≥ 50 and a base at $\leq 2\%$ in DNA but $\geq 10\%$ and at least 10 units in RNA. APOD no longer meets it; the other five still do.

PABPC1 remains unresolved

The new data retain G at chr8:100705591 in 61/470 names and C at chr8:100705604 in 73/430. All 61 G-supporting names also support C on the same qualifying aligned records; 56 of the original 65 shared names still support both. These linked observations must not be treated as two independent checks or converted into tumor mutations. Their origin remains unresolved.

Read context and remaining exceptions

Complete emitted records were preserved for all 3,426 preselected discrepancy-associated query names, including mates, secondary/supplementary placements and unmapped records. All names were recovered. New capture retained 9,061 named records, not merely alignments overlapping the questioned positions.

For APOD, 26 of 27 original C-supporting names now have qualifying G evidence at the site; the remaining name overlaps it but fails matched filters. For EXOC3, 14 of 22 original A names retain A; three now support C, one conflicts, two fail filters and two are unmapped. For MSH3/DHFR, 58 of 104 original G names retain G; 41 map without a base at this position, three fail filters and two are unmapped. These are changed alignment/filter outcomes, not proof of corrected origin.

Other original exceptions remain in the record

Group / locus	Original RNA	New RNA
WWC1	C3, T21	T21
RGS14	C27, G4	C25, G6
PTCH1	A22, T4	A21, T1
NOTCH1	A22, T17	A23
MYO18A	C3, G23	C3, G23
NF1	G5, T17	T23
SEPTIN9	G7, T27	G7, T27
MAPQ60-only: FANCD2	C72, T153	C73, T155
MAPQ60-only: SDHAP2 / MUC20-OT1	A913, G357	A889, G326
MAPQ60-only: FOXD4L4	A22, G70	A21, G69

The three MAPQ60-only flags were originally produced by tightening the DNA mapping-quality filter; they remain separate sensitivity findings. This table shows their matched baseline-filter RNA counts. MLLT3 chr9:20414378 is newly below 90%: original A24/G2 becomes A24/G3 (depth 27), below the strong screen's depth requirement.

Methods and verification

Matched count filters require MAPQ ≥30, BQ ≥25, proper pairing and exclusion of flags 0xF0C. Qualifying A/C/G/T observations are collapsed by exact query name; conflicting bases are excluded. Coverage means qualifying A/C/G/T, not reference skips. Capture preserves all SNP-base-overlapping records plus every emitted record for frozen names; D/N-only bookkeeping for other names is incomplete by design.

An independent aligned-pair implementation reproduced all 4,015 per-base count rows and material exception transitions. The capture logic passed synthetic boundary/flag/name controls; final BAM quick-check and complete input-pair counts passed. The complete record and per-name tables, scripts, frozen design and audit receipts are retained in the evidence archive.

Original Caris STAR 2.7.8a/CTAT/two-pass settings differ from local STAR 2.7.11b/primary-genome/D8/one-pass settings, including overlap, splice and chimeric parameters. This is not a controlled sparsity-only experiment, a validated identity/contamination assay or a clinical sensitivity test. New placements are not automatically corrections. The clean-read sensitivity and all 16 original review sites remain available, including coverage losses.

Local full pass completed in 26.8 minutes. The selective BAM is 64.1 MB; monitored peak main-pipeline RSS was 8.40 GB and minimum free disk 4.99 GB. No full BAM, new clinical target or external outreach resulted. The biological v2 handoff remains unchanged.